

IRC and forums

Two common places where cyber criminals group and share information. E-mail lists and working credit card numbers are traded most often

Botnets

A group of remotely controlled computers that can together be used for a number of malicious purposes

Not about being cool

The biggest draw for hackers today is not a show of coding prowess, but a simple urge to earn money

located in a business hub for example, is a very good target for such attacks.

Credit card details cannot be changed after a stint at a Cybercafe, and is something that should never be keyed in on a public computer. If it is absolutely essential, a simple precaution is to use an on screen keyboard. Go to the Start > Accessories > Accessibility > On-Screen Keyboard on any Windows system. A virtual keyboard will show up on the screen, and you can use the mouse to key in any sensitive information. Many banks offer virtual keyboards as part of their websites, use these for security when available.

A virtual keyboard is just an accessibility tool, and the default virtual keyboard can bypass only hardware keyloggers. If you frequent cybercafés, and are an advanced user of on-screen keyboards, there is a software called on-screen keyboard portable that can be used across machines. You can customize the keyboard, and take the settings with you. Users, can for example, set hover preferences that keys in the text without resorting to mouse clicks.

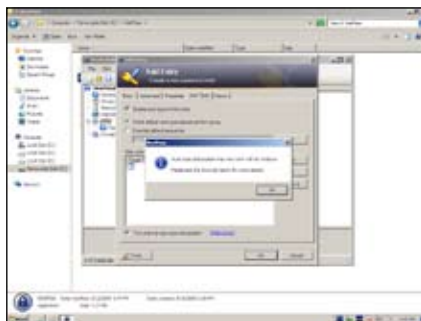
Keypass password safe

Not all software keyloggers can be bypassed with the use of virtual keyboards, as the keyloggers log the keypresses from the virtual keyboards as well. Some keyloggers are even configured to log only details from on-screen keyboards, which makes the work for the malicious person much easier, as the un-sensitive keypresses from the keyboard are not logged at all. The safest way to bypass this, is to use a process called obfuscation. Obfuscation basically allows keyloggers to log a certain combination of keys, while keying in a different combination. There are some programs that are targeted at different obfuscation algorithms, and bypass them, but this still remains the most secure approach to keying in sensitive details. The user does not have to enter the keys manually, or use a virtual keyboard in this process. The password, login details, and other sensitive information is stored in a secure computer, then the program is run on a public machine. At this point, the program automatically enters the details, while obfuscating the characters used.

The best software for this is an open source program called KeyPass. Open source software can be trusted in matters of security at least, since a lot of people stand to lose their data, the code is open for everyone to examine. That does not mean that the software is easy to bypass, quite

the opposite, the open scrutiny gives more credence to the software.

KeyPass is a password "safe", that stores all your passwords and sensitive information in one encrypted database file. This database file cannot be easily accessed, and can be taken around with you on a USB drive. You need



Obfuscating the text while auto-keying

just one password to access the database. To keep matters simple, people tend to use the same password across all the websites they access. If one account is compromised, all of them are. The best way to keep all your accounts secure is to keep different passwords for them, and change the passwords regularly.

People who do this have a code or a system, where that they use to choose and cycle the passwords for different sites. Despite that, it is a good idea to use a password safe to store all your passwords. This way, you have to remember just one strong password, and you don't have to key in your passwords on any site.

KeyPass works by using a sensitive information database. There are some steps that KeyPass takes to protect this database from intrusion. When you first run the software, you will be asked to create a master key and a keyfile. The keyfile is a file with random data, that is many times more secure than a password. A password more than 16 characters long for example, becomes very difficult to remember. Therefore, crackers or brute force methods, that exhaust all the possible combinations can get into the information with relative ease. In practice, this takes a long time, but is not impossible. A keyfile makes intrusion much more difficult, and beyond the purview of current brute-force intrusion methods. First you will have to create a master password for the database. The longer the password, the stronger it is. The more numbers and special characters you use, the better the password. KeyPass

gives you an indication of how strong your password is. Anything over 50 bits in the "estimated quality" field is good to go.

You can choose to opt out of creating the keyfile. However, if you are storing a lot of sensitive data in the database, it is a good idea to create a keyfile. Click on Create to make a keyfile. There are two approaches to make a keyfile. Both these approaches ensure that the keyfile is truly random, and not pseudo-random. On the left is a field of noise, which the user has to run over with a mouse. Alternatively, a user can choose to key in random data from the keyboard. The more secure approach is to run the mouse randomly over the noise field, as even a random set of keys from a human user concentrates on a few keys, and is not that random.

You can stop once 256 characters are reached, which is many times more secure than a 12 character password. Now the database should have a few default fields. Fill this in and add details as necessary. For all the frequently used entries, you need to enable auto-type. Auto-type automatically fills in the username and passwords in a number of sites. Auto-type works for most kinds of windows, but not all. Any field in a standard web page is accessible, and all the browsers are supported. There are a few places where auto-type does not work, such as a command line interface. Unless you are using a text-based browser, this should not be a problem.

When you edit the preferences for an entry, go to the Auto-type tab, and check Enable auto-type for this entry. Also check two-channel auto-type obfuscation. This is the feature that makes the text invisible to any kind of program that discovers cloaked text, or software keyloggers. You will be warned that auto-type obfuscation will not work everywhere, but ignore this warning as it is applicable to only the command-line type scenarios.

Carry your browser with you

Portable Firefox, and Opera@USB are both browsers that you can take with you to cybercafés. A simple way to bypass the process of entering e-mail information is to select the "remember me" option on a secure computer. The browser does not save any data on the host system, and all the necessary files are stored on the portable memory. Since the programs are already installed, you don't need administrative privileges to run the software. Additionally, this will save you a lot of time, as your favorites and bookmarks are carried with you in the browser. To use either of these browsers, just use the installer and install the browser in the relevant USB drive.

SMART TIP

It is a good idea to change all the passwords you used on a public computer, once you access your account or profile from a secure computer